Research design: from question to evidence

Each research question is tied to a concrete method, artifact, and the metric its conclusion rests on

Research question	Method	Artifact / data	Primary metric
RQ1 tail-latency predictability	load sweep to overload, interleaved repeats	benchmark harness; HDR latency histograms	**p99 / p99.9 tail & jitter vs load**
RQ3 safety-model overhead	same sweep, resource capture	getrusage samples per run	**peak RSS, CPU, throughput**
RQ2 defect prevention	controlled fault injection	defect catalogue; ASan/TSan + rustc logs	**defect-class taxonomy** compile / runtime / missed
RQ4 migration cost	FFI migration prototype	C++ host + Rust C-ABI seam	**seam size, unsafe scope, per-call cost**

RQ2 & RQ4: **final** (environment-insensitive)

RQ1 & RQ3: **preliminary** (await bare-metal testbed)

All four answered from one matched pair of C++/Rust implementations built from a single frozen specification.